

Kelowna, Canada, July 8, 2025

Dear Editors,

We are pleased to submit our manuscript entitled “Spatial Ecology of Jaguars in a Coastal Amazon Island Ecosystem” for consideration as a research article in PLOS ONE.

This manuscript presents the first detailed investigation of jaguar spatial behaviour in an insular Amazonian setting. Using fine-scale GPS tracking data from seven individuals on Maracá-Jipioca Island—a low-disturbance, prey-rich protected area—we examine patterns of home-range dynamics, movement, and activity across habitat types and tidal cycles. Despite strong spatial constraints imposed by the island environment, we find that jaguars retain many of the territorial and sex-specific space use patterns typical of mainland populations, while also exhibiting novel adaptations, including pronounced diurnality and coastal habitat preferences modulated by tidal variation. Our findings highlight how spatial structure, prey base, and disturbance interact to shape movement strategies in large carnivores. They also contribute to the broader understanding of predator ecology in fragmented or isolated systems, and may inform conservation strategies in both island and mainland reserves as they become increasingly shaped by environmental change.

We believe this work is well suited to PLOS ONE, given the journal’s emphasis on rigorous methodology, transparent reporting, and relevance to both conservation and animal ecology. In addition, we note that all data and analysis code required to reproduce this study are openly available at https://github.com/QuantitativeEcologyLab/island_jaguars, and we confirm that there are no conflicts of interest to declare.

Thank you for your time in considering our submission, and we look forward to your response.

Kind regards,

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